BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 007 · Compression and Storage Calculations

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What does a sample rate of 8,000 Hz mean?

Q2. How many bytes are in 96 bits?
A. 24 bytes
B. 96 bytes
C. 12 bytes
D. 768 bytes

Q3. A text file contains 240 characters. The stated encoding uses 8 bits per character. Calculate its raw size in bits and bytes.

Q4. (a) A bitmap uses 5 bits per pixel. How many colours can it show? What if you add one bit? (b) A 50 × 20 pixel image doubles in width and height. Colour depth stays the same. How do pixel count and raw size change? Explain.

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I Do · Read new knowledge

Key vocabulary

compression — Reducing the number of bits needed to represent a file.

lossless — Compression that allows the original to be restored exactly.

lossy — Compression that permanently removes some original data.

file size — The quantity of storage required for a file.

New learning

Why compress files?: Compression reduces the amount of storage needed and can reduce transmission times.

Lossless: Lossless compression makes files smaller while allowing exact reconstruction of every original bit. Use it for source code and important documents.

Lossy: Lossy compression permanently removes some data. It can reduce photos, music or video file sizes when some loss of quality is acceptable.

Calculating savings: Storage saved = original size − compressed size. Keep units consistent; if asked for a percentage, divide by original size and multiply by 100.

Worked example

A text program and a photograph must be compressed. Choose a method for each and calculate the saving if a photo falls from 800 KB to 500 KB.

1. Source code must remain exact, so choose lossless compression.

2. A website photograph may use lossy compression if some detail loss is acceptable.

3. Subtract compressed size: 800 KB − 500 KB = 300 KB saved.

4. Percentage saved = 300 ÷ 800 × 100 = 37.5%.

We Do · Practise with guidance

Guided 1

Should a spreadsheet containing exact examination marks use lossy or lossless compression? Explain.

Worked method: Lossless: all original data must be recoverable exactly.

Guided 2

A 900 KB image compresses to 600 KB. How many KB were saved?

Worked method: 900 − 600 = 300 KB.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed:

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Sample rate is samples per second. At 8,000 Hz, 8,000 samples are taken each second. A higher rate can capture more detail but uses more space.

Q2. C — 12 bytes

Q3. 1,920 bits and 240 bytes.

Q4. (a) 32 colours; one extra bit doubles this to 64. (b) 1,000 pixels become 4,000. Raw size is four times larger: both dimensions double.

I Do / We Do pages · Guided practice

Q1. Lossless: all original data must be recoverable exactly.

Q2. 900 − 600 = 300 KB.

Q3. 300 ÷ 900 × 100 ≈ 33.3%.

You Do pages · Code First

```
original = 1000
compressed = 600
print(original - compressed)
```

Expected result: 400

You Do pages · Independent questions

Q1. It requires less storage or may transfer faster [1].

Q2. The original file must be recovered exactly [1], because losing any original instructions could corrupt the program [1].

Q3. 12 − 9 = 3 MB saved [1]. Lossy may be suitable [1] because some sound information can be discarded to reduce file size [1], provided quality remains acceptable for the purpose [1].

Exit Ticket · Model answer

True

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BLACK PEN Answer all questions first. GREEN PEN Then: tick and rewrite mistakes.

Guided 3

What percentage of the original file size was saved?

Worked method: 300 ÷ 900 × 100 ≈ 33.3%.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Code First · Predict → Write → Test → Improve

Change compressed to 600. What prints?

```
original = 1000
compressed = 700
print(original - compressed)
```

write

Question 1

Identify one reason to compress a file. [1 mark]

Question 2

Explain why lossless compression is appropriate for computer program files. [2 marks]

Question 3

A sound file falls from 12 MB to 9 MB after compression. Calculate storage saved and justify whether lossy may be suitable for online listening. [4 marks]

Exit Ticket · Answer before checking

Lossless compression can reproduce the exact original data.

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